

NEBRASKA CITY, NE, USA

WMO: 725541

Lat: 40.606N

Lon: 95.864W

Elev: 351

StdP: 97.17

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 04993

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.8	-15.6	-22.5	0.5	-16.5	-20.5	0.6	-14.1	13.7	-2.0	12.2	-1.9	3.7	330	0.549

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	34.0	24.1	32.5	23.7	31.2	23.3	26.4	31.0	25.4	29.9	24.5	29.0	5.4	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.1	21.1	29.2	24.0	19.8	28.0	23.0	18.5	26.9	84.4	30.6	80.0	29.8	75.8	29.1	29.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.9	10.7	9.2	-22.1	36.8	2.4	1.5	-23.8	37.9	-25.2	38.8	-26.6	39.6	-28.3	40.7		
(5)			-22.5	28.0	2.4	0.9	-24.2	28.6	-25.6	29.2	-26.9	29.7	-28.6	30.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.5	-3.3	-1.8	5.8	11.5	17.5	23.0	24.6	23.2	19.8	12.5	5.4	-1.4		
	DBStd	11.20	6.65	6.73	6.63	5.49	4.87	3.21	3.13	2.95	4.28	5.20	5.70	6.02		
	HDD10.0	1503	414	333	164	48	4	0	0	0	32	156	353			
	HDD18.3	3142	672	565	389	212	76	3	1	2	33	191	387	611		
	CDD10.0	2043	0	2	35	92	236	390	453	409	294	111	19	1		
	CDD18.3	639	0	0	2	6	49	143	196	153	78	11	0	0		
	CDH23.3	5872	0	0	23	89	447	1340	1909	1267	681	112	3	0		
	CDH26.7	2136	0	0	3	19	133	495	775	451	233	28	0	0		
(14) Wind	WSAvg	4.3	4.6	4.7	5.1	5.4	4.7	4.0	3.2	3.0	3.7	4.3	4.7	4.5		
(15) Precipitation	PrecAvg	833	22	29	49	77	131	131	114	73	73	60	45	26		
	PrecMax	1327	37	89	105	185	300	244	483	204	157	180	96	76		
	PrecMin	639	3	2	0	17	39	66	17	13	15	2	1	0		
	PrecStd	164	11	22	33	38	72	48	98	44	41	42	26	20		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.7	19.8	26.2	29.3	33.2	35.0	37.2	34.8	33.6	30.1	23.0	15.5		
		MCWB	7.0	10.1	16.5	18.1	20.5	23.8	23.8	25.6	21.8	20.1	13.2	9.5		
	2%	DB	11.2	14.8	22.8	26.1	30.0	32.9	34.3	32.7	31.5	26.3	19.8	12.1		
		MCWB	5.9	8.8	13.3	15.7	20.1	23.0	24.9	24.5	22.0	17.4	12.7	8.1		
	5%	DB	7.8	10.8	19.7	23.0	27.8	31.4	32.6	31.1	29.7	23.8	16.8	9.3		
		MCWB	4.2	6.3	12.6	14.4	18.6	22.9	24.7	24.3	21.4	16.5	11.3	6.0		
	10%	DB	5.2	7.4	16.3	20.8	26.0	29.8	31.0	29.1	27.4	21.3	14.0	7.0		
		MCWB	2.7	4.2	11.0	13.5	18.0	22.1	24.2	23.1	19.7	14.7	9.1	4.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.5	12.1	17.9	19.1	23.2	26.7	28.0	27.5	25.0	22.5	16.1	12.5		
		MCDB	12.8	16.4	22.9	26.1	28.3	31.8	32.4	32.7	29.3	26.9	19.6	13.5		
	2%	WB	6.3	8.5	16.0	17.5	21.8	25.2	26.7	26.1	23.8	20.3	13.9	8.5		
		MCDB	10.3	15.3	20.2	23.2	27.2	30.2	31.3	30.3	27.7	24.3	17.3	10.9		
	5%	WB	4.5	6.4	13.8	16.0	20.6	24.1	25.7	25.0	22.8	18.0	11.9	6.1		
		MCDB	7.8	10.5	17.8	21.3	25.6	29.0	30.2	29.3	26.7	21.4	16.1	9.3		
	10%	WB	2.8	4.1	11.1	14.5	19.3	23.2	24.8	24.0	21.7	15.6	9.7	4.3		
		MCDB	5.3	7.1	16.9	19.4	24.1	27.9	29.4	28.1	25.9	19.9	14.3	6.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.5	9.7	11.2	11.6	10.9	10.5	10.3	10.2	11.6	11.4	10.8	9.0		
		MCDBR	13.5	15.4	16.1	15.3	13.2	12.2	12.1	12.2	13.4	14.6	14.9	13.0		
	5% WB	MCWBR	9.1	9.4	8.5	7.9	5.9	5.3	5.3	5.7	5.4	7.4	8.4	9.1		
		MCWBR	12.5	14.3	13.1	13.2	11.2	10.8	10.3	10.6	11.0	11.9	13.5	11.9		
(40) Clear-Sky Solar Irradiance	taub		0.276	0.311	0.324	0.377	0.407	0.413	0.420	0.416	0.373	0.331	0.297	0.282		
	taud		2.487	2.399	2.423	2.294	2.278	2.308	2.282	2.308	2.382	2.485	2.510	2.475		
	Ebn at Noon		897	904	930	895	872	864	854	847	865	871	862	858		
	Edh at Noon		75	95	104	127	132	128	131	124	108	87	73	70		
(44) All-Sky Solar Radiation	RadAvg		2.11	3.00	3.91	4.94	5.83	6.50	6.50	5.69	4.70	3.32	2.33	1.77		
	RadStd		0.16	0.19	0.35	0.31	0.48	0.48	0.44	0.30	0.31	0.37	0.18	0.18		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	+1.20	N/A	N/A	N/A	N/A
(47)	Regional (42 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page